Progress Report 3

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Project: Riipen EMS Data Analysis and Forecasting

Repo: https://github.com/dondemici/W26_4495_S2_DundeeA/

Work Log

Week	Date	Number of Hours	Description of Work Done
1	18-Jan-26	0.5	Meeting with Priya, initial discussion of topics
	21-Jan-26	2	NEMSIS Data initial proposal
	23-Jan-26	0.5	Meeting with Solaris, Tony Tsui
	24-Jan-26	1	Initial works on the final proposal, intro and proposed research project
2	25-Jan-26	2	Align Intro and Proposed Research Project on the final proposal – Riipen link finally given
	26-Jan-26	4	Final works on the final proposal – Estimate and update timelines, build contract, update references, update AI usage table and prompts and perform final review and revisions
	30-Jan-26	2	Review of NEMSIS website, submission of data request for the project, communication of updates to Tony Take Datacamp course on Introduction to Databricks, Setup a community edition Databricks account,
	31-Jan-26	2.5	Review data sent by NEMSIS, extract data (about 2 hrs. processing to extract 200GB data), build initial python code to review data
3	01-Feb-26	2	Update project plan to Agile format, reconsider performing a subset of data performing activities of the whole project in a shorter span of time, create ems eda notebook
	03-Feb-26	2.5	Review data dictionary form NEMSIS and study the headers in the raw dataset. Align equivalent mapping in the data dictionary to get more meaning.
	05-Feb-26	2.5	Provide updates to Riipen stakeholder, further revise project plan. Update ems eda notebook
	06-Feb-26	1.5	Provide updates to Priya, consolidate and review all generated files including notebook, upload to github
	07-Feb-26	1.5	Build end-to-end notebook pipeline from EDA to initial model application and evaluation

Week	Date	Number of Hours	Description of Work Done
4	08-Feb-26	2.5	Update notebook pipeline, troubleshoot time data format, review next steps for the succeeding week.
	10-Feb-26	2	Filter Jan2024 from PCR_Events table using MS SQL SM.
	12-Feb-26	2.5	Analyze Exposure data. Join with PCR Events – results: Null Values for eOther for Jan2024.
	14-Feb-26	3	Review count of YES/No records for Exposure data. Check hospital admit time of Yes data – results: Not Applicable.
5	15-Feb-26	2.5	Revisit hypotheses and pivot. Analyze FACTPCR dataset. Focus on 20K records, eg weekly exposure forecast using SARIMA or Prophet.
	16-Feb-26	2	Design dashboard for forecasting app. Ask forecaster friends some features and functions that are ideal for forecasting app.
	17-Feb-26	2.5	Initiate and code Streamlit app. Include sidebar settings and main context for dashboard.
	18-Feb-26	2	Reach out to NEMSIS report for additional provided injury data –got response that they don't have the data readily available. Build script for video. Finalize Streamlit app.
	19-Feb-26	3	Shoot and edit video
	20-Feb-26	3.5	Finalize video – add in slides, Initiate midterm report, update readme file
	21-Feb-26	2	Review and polish midterm checklists. Update AI Usage table and List of Prompts table. Upload video in youtube.
6	22-Feb-26	1	Finalize and upload midterm report to Blackboard (initial submission)
	23-Feb-26	2.5	Performed more works today using PCRTime data to extract historical daily exposure data as time series to forecast using 7-day moving average which current serves as best model compared to naïve and exponential smoothing.
	24-Feb-26	1	Preparation of presentation to employer and planning of next steps
	25-Feb-26	1	Employer session on project updates
	26-Feb-26	2.5	Initial build of incorporating data pre-processing in the app, shift from fake data to real data
	27-Feb-26	1	Midterm check-in. Continue incorporating SQL code connection in the app
	28-Feb-26	2	Incorporation of forecasting models into the app
7	01-Mar-26	1.5	App testing of data connection and the forecasting models, capture of issues
	02-Mar-26	1	Progress Report 2 update and submission.
	03-Mar-06	2	Include additional forecast models for seasonality – Holt & Holt-Winters. Include error handling for Holt-Winters if available data is only 1 year.

Week	Date	Number of Hours	Description of Work Done
	04-Mar-06	1	Request of additional data, cover more years from 2020-2023
	05-Mar-06	2	Sending of weekly updates to client.
	06-Mar-06	2.5	Week 7 check-in, review data given and follow up additional needed data.
	07-Mar-06	2	Initial code works on connecting to AI API to generate recommendation section.
8	08-Mar-06	2	Review data given and follow up FACTPCRTIME data since it still not available in the 2nd batch of files provided. Review requirements for hosting in Streamlit.
	09-Mar-06	1	Progress Report 3 update and submission.

Summary Description

This week, I expanded the forecasting component by integrating additional seasonality models (Holt and Holt-Winters) and implemented error handling for limited yearly data. Additional historical data from 2020–2023 was requested and partially received. I coordinated weekly updates with the client, reviewed newly provided datasets, and flagged missing FACTPCRTIME records for follow-up. Development also advanced with initial integration of an AI API for generating recommendation sections and planning for deployment via Streamlit. The Week 7 check-in was completed, and Progress Report 3 was submitted on time.

Key Activities:

- Added Holt and Holt-Winters models for seasonality; implemented handling for one-year data gaps.
- Requested extended dataset (2020–2023) to improve forecast accuracy.
- Sent weekly progress update to the client.
- Conducted Week 7 review and data validation with noted missing files.
- Began AI API connection for recommendation generation.
- Reviewed hosting setup and requirements for Streamlit deployment.
- Submitted Progress Report 3 summarizing current progress and next steps.

Repo Check-in

I have committed my work daily to the shared GitHub repository, focusing this week on adding more forecasting models, requesting additional year data and building initial AI API works for recommendation generation. The main implementation artifacts are under the Implementation directory, and documentation is stored under Documents and Reports. ie proposal and the progress reports.

Folder File	File	Brief Description
DocumentsAndReports	DundeeA_Proposal.pdf	Original project proposal outlining goals (EMS high-strain demand forecasting and injury-risk context), data sources (NEMSIS), methods, and initial timeline.
	DundeeA_ProgressReport1.pdf	First progress report summarizing early work, issues encountered, and initial plan adjustments.
	DundeeA_MidtermReport.pdf	Midterm report describing project status, key findings, and feedback from the employer check-in.
	DundeeA_ProgressReport2.pdf	Second progress report detailing recent app integration, forecasting work, and SQL connectivity updates.
	DundeeA_ProgressReport3.pdf	Third progress report detailing addition of forecasting models, request of additional data and initial build of AI API.
	CSIS 4495 002 Applied Research Working File.xlsx	Working file tracking tasks, milestones, timelines, and evaluation components for the project.
Implementation/1 EDA	EMS_EDA.ipynb	Initial exploratory notebook examining NEMSIS schema, value distributions, and key variables for EMS demand and exposure analysis.
	ems_eda2.ipynb	Follow-up EDA notebook refining feature selection and exploring additional patterns in the EMS incidents data.
	ems data review.ipynb	Notebook reviewing EMS data quality, cleaning steps, and suitability of fields for modeling and forecasting.
	ems_data_review databricks.ipynb	Databricks-based version of the EMS data review pipeline for scalable EDA and transformation.
	PCREvents.ipynb	Notebook focusing on PCR events data to inspect event-level fields and relationships relevant to exposure and high-strain events.

Folder File	File	Brief Description
	nemsis_sprint0_3_sample.csv	Sampled and consolidated subset of the NEMSIS dataset used for initial exploration and testing.
	NEMSIS Header Table.xlsb	Header reference file describing NEMSIS data elements to support meaningful interpretation during EDA.
	sample_DISPATCHDELAY.txt	Raw sample dispatch delay file used to test parsing, filtering, and integration logic.
	sample_Pub_PCREvents.txt	Raw sample public PCR events file used to validate event-level ingestion and filtering.
	sample_TURNAROUNDDELAY.txt	Raw sample turnaround delay file for testing delay-related feature extraction.
	SQLFilter1	SQL script defining the EMSData database and core table structure for storing filtered EMS data.
	PCR_Exposure_Minimal (SQL script)	SQL script to create the PCR_Exposure_Minimal table with a reduced set of key exposure-related fields.
	PowerShellJan24Filter	PowerShell script used to filter large NEMSIS event files by date for manageable EDA input.
Implementation /Modeling	Modeling.ipynb	Notebook implementing the initial end-to-end EMS forecasting models, including feature selection, model training, and evaluation for high-strain demand.
Implementation /Product	.gitignore	Git ignore file configured to exclude Streamlit secrets and other sensitive or local configuration files from version control.
	ems_forecaster_appv1.py	Initial version of the EMS forecaster app implementing basic UI, data loading, and simple forecasting workflow for EMS demand.
	ems_forecaster_app.py	Updated EMS forecaster app integrating the latest forecasting models and improved logic for interacting with real EMS data.
	ems_forecaster_check.ipynb ems_forecaster_check.py	Sanity check of actual data using the various forecating models.
	requirements.txt	List of requirements of the Streamlit app.
Misc	EMS Forecast Call Lifecycle.pptx	Slide deck outlining the end-to-end lifecycle of an EMS forecast call from data ingestion to model output and use.
	EMS Forecasting App.pdf	Visual overview document describing the EMS forecasting app screens, workflow, and key features.

Folder File	File	Brief Description
	EMS Forecasting Midterm.pptx	Midterm presentation slides summarizing project motivation, data pipeline, models, and interim results.
	NEMSIS_Data_Request	Details of data request for 2020-2023
	SD	Systems diagram file illustrating the architecture and data flows between EMS data sources, models, and the app.
	VideoLink	Text file containing the link to the project demonstration or presentation video.

AI Use Section

AI Tool Name	Version Account Type	Specific Feature for which the AI Tool was Used	Value Addition *(What value did you add over and above what AI did for you?)*
Perplexity	Education Pro	Asked AI to surface literature that would support framing the problem that demonstrates occupational hazard encountered EMS personnel	Having done a coop role in VCH, I learned that work-related injuries are handled by WorkSafeBC so I also asked AI to check for any statistics coming from the organization, this is on top of the 2 it produced on CUPE and JEMS
Perplexity	Education Pro	Asked AI to confirm if there's any AI or ML related studies or projects published using the NEMSIS data in order to check uniqueness of my study	Gave input to AI about the focus of my studies which is really about forecasting work-related injuries for emergency personnels so that it serves as comparison for other studies that may have been done
Perplexity	Education Pro	Asked AI to evaluate my timeline if reasonable and for any crucial activities I may have missed	Provide the milestones I crafted as well as the activities envisioned so that AI provide further input on anything I missed
Perplexity	Education Pro	Asked AI to help build and code the Streamlit EMS forecasting app.	Provided the requirements from selecting models like SARIMAR or prophet, week duration, special features like taking into account concerts and more as well as driving how the main content look like with bargraphs and recommendation.

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Perplexity	Education Pro	Asked AI to review SQL codes particularly in troubleshooting the time format of the hospital admit time.	Discovered that with my SQL codes, no data is being filtered out so investigated why and found that time format is somehow messing up the filter so I asked AI 3 scenarios on the format.
Perplexity	Education Pro	Asked AI to refine the video script I created, improve the flow and make it more concise so that it's at least 6 minutes long.	Created the outline and initial video script, discussing the major contents I would like to be present in the video then feed it to AI for further improvement.
Perplexity	Education Pro	Asked AI to provide a crash tutorial on Streamlit – app to use and how it works	Provided requirements to build an app and to have dashboard. AI is able to narrow down to Streamlit as a platform that's able to meet my requirements.
Perplexity	Education Pro	Asked AI to provide options for apps that can process large dataset aside from databrick, given that the app is only able to handle 5-10gb data.	Provided the context that I would like to go with Databricks but given its limitations, I would like to explore more app.
Perplexity	Education Pro	Asked AI to layout a couple of approaches for me in terms of joining large dataset in Jupyter Notebook.	Provided context that I'm able to use SQL to narrow down first dataset but given I'm joining to other large dataset, Jupyter Notebook code kept on crashing.
Perplexity	Education Pro	Asked AI to build data connection SQL codes and incorporate into the app	Provided context and initial codes built in order for AI to enable the connection of the app to the SQL data
Chagpt	Free Account	Asked AI about how to properly store confidential credentials so that it's not uploaded in github	Provided situation where I have an app code that connects to SQL and would need to hide the credentials
Perplexity	Education	Asked AI to build actual forecasting models and incorporate into the app	Listed the models to be used and specified where to incorporate into the app

AI Tool Name	Version Account Type	Specific Feature for which the AI Tool was Used	Value Addition *(What value did you add over and above what AI did for you?)*
Perplexity	Education	Incorporate additional forecasting models into the app	To ensure robustness of forecasting models, prompted AI to add 2 more models.
Claude	Free Account	Build code to use LLM to generate recommendation	Gave specific input to refer to the forecast results and take those as input to generate recommendation output.

List of AI Prompts

Prompts
Given my project and objective on analyzing EMS data and forecasting information to improve QA protocols to lessen work-related injuries of emergency personnel, find relevant literature that would support demonstrating the occupational hazards encountered by these workers.
When the literature was given, I further prompted if AI can give me stats by WorkSafeBC, having learned about the organization during my coop role time in VCH.
Given that I am doing a project on NEMSIS data, provide confirmation on any studies done in the past that is related to AI or ML so that I understand the significance and uniqueness of my study. My study focuses on forecasting work-related injuries for emergency personnels.
Review and evaluate my timeline of the project from EDA, cleaning, model evaluation and selection until dashboard reporting and check if reasonable given the 10-12.5 hrs I can allot weekly for a 2-month period. Mention any activities I may have missed to include.
Help me design and code a Streamlit app for an EMS exposure-risk forecasting dashboard where the user can choose SARIMA or Prophet, select the forecast horizon with a slider, optionally toggle scenario flags (concerts, political rallies), and then see historical plus forecasted weekly exposure counts with prediction intervals.
Review and debug my SQL code for joining NEMSIS tables on PcrKey and converting hospital admit time / PCR completion time fields into a consistent datetime format that I can use for weekly aggregation.
Refine the video script I created for my midterm presentation on EMS exposure forecasting—improve the flow, make it more concise, and keep the runtime around 6 minutes while preserving key technical details.
Provide a crash tutorial on building a Streamlit app for data science projects: how to structure the main script, add sidebar controls, display plots and tables, and run the app locally.
Suggest options for tools or platforms that can process large datasets besides the free Databricks edition, given that it can only handle around 5–10 GB, and explain pros/cons for my NEMSIS project.
Lay out a couple of approaches for joining large relational datasets in Python/Jupyter (e.g., from NEMSIS): when to rely on pandas merges vs doing joins in SQL first, and how to manage memory efficiently.

Prompts
Have a Python-based EMS forecasting app and I need to connect it to a SQL database instead of using fake data. Here is my current app code and data structure—please write the SQL connection code (including queries) and show me exactly how to integrate it into my app so it can read real EMS data.
Have an app that connects to a SQL database, and I'm pushing the code to a public GitHub repo. How can I properly store and hide confidential credentials (server, username, password, connection string) so they are not committed to GitHub, and what is the recommended way to load them safely at runtime?
Building an EMS demand forecasting app and want to integrate the following models (e.g., SARIMA, Prophet, Naïve, Moving Average and Exponential Smoothing). Here is my current app structure and data pipeline—please write the model training and prediction code and show where to insert it so the app can generate and display forecasts based on the selected model.
Add two additional forecasting models for time series seasonality — Holt's Linear Trend and Holt-Winters methods. Implement automatic selection of the best model based on AIC or RMSE. Include error handling for Holt-Winters if only one year of data is available
Use the forecast results as input data. Generate a recommendations section that summarizes key performance trends and suggests actions based on predicted values. Structure the output in a brief, client-friendly format suitable for report inclusion